

Michael Beals

Stanford, California | mbeals27@stanford.edu | <https://bealsmichael.github.io>

EDUCATION

Stanford University | Stanford, CA

GPA: 3.69

BS Mechanical Engineering | Graduation: June 2027

RELEVANT EXPERIENCE

TSMC Arizona | Equipment Engineering Intern, Etch (ETC3) | Phoenix, AZ

June 2026 – August 2026

- Applied PID control principles to analyze the motion profile of the wafer transfer robot arm, evaluating movement behavior to extend overall arm service life.
- Characterized lubricating greases using material property data and rheological testing (squeeze flow) to compare flow and film behavior and improve lubricant selection for the transfer arm.
- Analyzed airflow and pressure differential across the fan filter unit and air curtain of the tool mini-environment to assess flow uniformity and containment.
- Advanced to the semifinals of TSMC's company-wide intern project competition.
- Collaborated daily with process engineering and counterpart teams in Taiwan, working across languages and time zones to align on tool behavior and analysis methodology.

CHARM Lab | Mechanical Engineering Research Intern | Stanford, CA

June 2025 – March 2026

- Designed and tested Tension Reduction Devices (TRDs) for the Vine Robot, a soft robotic system for confined-space exploration, integrating modular tip mounts and TRD assemblies to reduce resistive forces.
- Modeled and fabricated TRD prototypes using Onshape and Fusion 360 with 3D printing, ultrasonic welding, and soldering.
- Developed Teensy microcontroller modules in Arduino IDE for synchronized multi-device operation.
- Built a transparent in-pipe test setup and modular test courses to evaluate TRD performance and robot mobility under varied conditions.

Stanford Space Initiative | Rockets Team Member | Stanford, CA

September 2024 – December 2024

- Reviewed and documented Python flight control code for the Fountain Hopper rocket, mapping module purpose, signal flow, and interfaces to improve team onboarding.
- Coordinated with Hardware and Propulsion teams to align software interfaces with sensor and actuator constraints.

PROJECTS

Narsil Bottle Opener | Individual Design & Manufacturing Project (ME 103)

Winter 2026

- Designed a functional cast-aluminum bottle opener end-to-end in Fusion 360 as sole designer, iterating the CAD model across foam and machined plastic prototypes before committing to final geometry.
- Sand-cast the blade, cross guard, and lower handle in aluminum, engineering draft angles and pattern geometry so raised engravings reproduced cleanly in a single pour; registered two-part molds using cast-in alignment features in place of a pattern board.
- Turned the 6061 aluminum handle on a manual lathe with a radius cutter, and machined mating and assembly features on the mill and drill press using custom soft jaws and work holding.
- Salvaged out-of-spec hardware in-shop by die-threading, cutting, and belt-sanding oversized screws to fit, then wet-sanded and polished cast surfaces through progressive grits to a mirror finish.

CoralScapes | Coral Nursery Structure Optimization

Winter 2025

- Designed a submerged coral nursery frame for durability and manufacturability under real marine loading at 20 m depth.
- Ran FEA in Fusion 360 on aluminum 6061-T6 and stainless 316L to evaluate von Mises stress, displacement, and safety factors (1.45 to 4.99) under hydrostatic, drag, and buoyant loads, validating results against analytical hand calculations within 8%.

SKILLS & INTERESTS

Analysis & Design: CAD (Fusion 360, Onshape, SolidWorks), Finite Element Analysis, Thermal Analysis, PID Control, Rheological Testing & Material Characterization, Airflow/Pressure Analysis, Python, C++, Arduino IDE

Manufacturing: Sand Casting & Pattern Design, Manual Lathe (Turning, Radius Cutting), Milling, Drill Press, Workholding & Soft Jaws, Welding, Forging, Die Threading, Surface Finishing & Polishing, 3D Printing, Rapid Prototyping, DFM/DFAM

Interests: Study Abroad in Madrid, Spain | Club Soccer | Club Golf | Stanford Club Latino | Society of Latinx Engineers